Strategic Marketing and GARCH Stationarity under Jeffrey Updating

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Abstract

We study a two-player strategic investment model in which each player markets their return process to attract external capital by shifting the other's beliefs via Jeffrey conditioning — without altering the true return process. This marketing constitutes a precise form of lying: it adds virtual up-counts to a Beta-Binomial posterior, distorting both perceived drift and perceived volatility. The central finding concerns the structural source of GARCH volatility dynamics. We show that the marketing adjustment channel — absent under pure Bayesian updating — provides the mean-reversion force that converts an integrated GARCH process (IGARCH) into a stationary GARCH(1,1). Under pure Bayesian updating the persistence coefficient equals unity and the conditional variance follows a unit-root process; Jeffrey conditioning introduces a finite mean-reversion coefficient driven by optimal marketing spend. All GARCH parameters — the ARCH coefficient, persistence, GJR asymmetry, and baseline variance — are derived as explicit closed-form functions of model primitives: the binomial process parameters, the Beta prior, the performance fee, and risk aversion. A companion paper develops the full Stackelberg equilibrium and comparative statics.

JEL codes: D82, D83, G14, C11, C58

Keywords: Jeffrey conditioning, asymmetric information, GARCH stationarity, strategic investment, Beta-Binomial beliefs, mean reversion

1 Introduction

GARCH models are among the most widely used tools in empirical finance, yet their structural origins remain poorly understood. Persistence and mean reversion in conditional variance are typically imposed parametrically rather than derived from economic primitives. This paper provides a microfoundation: in a two-player strategic investment model with asymmetric information, GARCH volatility dynamics emerge from the interaction between strategic belief manipulation, optimal capital allocation, and a shared public history of returns.

The key finding is that the choice of belief-updating rule has structural consequences for the nature of the GARCH process. Under standard Bayesian updating, belief revision is driven entirely by hard evidence — observable returns with well-defined likelihoods. No mechanism exists for soft evidence such as marketing or persuasion. We show that this restriction implies a structural deficiency: the conditional variance of the resulting portfolio follows an integrated GARCH (IGARCH) process with unit-root persistence, which is both empirically counterfactual and theoretically unsatisfying.

Jeffrey conditioning (Jeffrey, 1965) allows beliefs to shift in response to soft evidence — evidence that changes credence without constituting a verifiable observation. In our model, marketing spend generates exactly this kind of shift: it adds virtual up-counts to the Bayesian posterior without changing the true return process. This is a lie in the precise economic sense. However, it introduces an endogenous strategic variable whose optimal level responds to the state of the world. This response constitutes the mean-reversion force that restores GARCH stationarity.

The contribution of this note is to establish this Bayesian versus Jeffrey distinction as a structural determinant of volatility dynamics — not a modelling detail. The result is clean: replacing Bayesian with Jeffrey updating converts IGARCH into stationary GARCH(1,1), with all parameters derived explicitly from economic primitives.

This complements the endogenous uncertainty literature (Kurz, 1994; Kurz and Motolese, 2001), which generates GARCH from the aggregate distribution of heterogeneous beliefs in competitive equilibrium. Our mechanism is bilateral and strategic rather than distributional and competitive. It also complements Kyle (1985), where informed trading converges to informational efficiency: in our model, belief manipulation via marketing converges to truth as accumulated history overwhelms the distortion. Both frameworks achieve asymptotic informational efficiency through different channels. The difference from Bayesian persuasion (Kamenica and Gentzkow, 2011) is that communication here is costly, updating is via Jeffrey rather than Bayes, and both parties simultaneously act as senders and receivers.

A companion paper (Rana, 2025) develops the full model: the Stackelberg equilibrium with closed-form solutions, comparative statics, the monopoly limit, and the convergence theorem showing truth dominates marketing asymptotically.

2 Model

2.1 Return Processes and Shared History

Two players $i \in \{1, 2\}$, with $j = 3 - i$, each own a risky process. Each period t, player i's process generates a return per unit of capital:

$$r_i^t = \begin{cases} u_i & \text{with true probability } p_i, \\ d_i & \text{with true probability } 1 - p_i, \end{cases} \tag{1}$$

where $u_i > d_i$ are publicly known constants and $p_i \in (0, 1)$ is the true up-probability, private to player i. Draws are i.i.d. across periods. True mean and variance:

$$\mu_i = p_i u_i + (1 - p_i) d_i, \qquad \sigma_i^2 = p_i(1 - p_i)(u_i - d_i)^2. \tag{2}$$

Both are determined entirely by p_i. Players have different true volatilities: $\sigma_1^2 \neq \sigma_2^2$ in general.

Both players observe the same public history of n past discrete returns from each process. Let k_i be the number of up-moves in player i's history. Player j's prior on p_i is $p_i \sim \mathrm{Beta}(\alpha_i^0, \beta_i^0)$. By conjugacy, the Bayesian posterior is:

$$p_i \mid \mathbf{r}_i \sim \mathrm{Beta}(\alpha_i^*, \beta_i^*), \qquad \alpha_i^* = \alpha_i^0 + k_i, \quad \beta_i^* = \beta_i^0 + n - k_i. \tag{3}$$

Since history is public, both players agree on this posterior. Let $S = \alpha_i^* + \beta_i^*$ denote the total pseudo-count; $S \sim n$ for large n.

2.2 Marketing as a Jeffrey Shift

The following describes player i's marketing acting on player j's beliefs; the symmetric case — player j's marketing shifting player i's beliefs about p_j — follows by exchanging i and j throughout. Player i spends $m_i \geq 0$ on marketing. This constitutes soft evidence in the sense of Jeffrey (1965): it shifts player j's credence over p_i without being a verifiable observation and without altering the true process (μ_i, σ_i^2). The Jeffrey rigidity condition holds since $\Pr(r_i^t = u_i \mid p_i) = p_i$ is determined by the true process, not by marketing. Formally, marketing generates a pseudo-count shift with diminishing returns:

$$\delta(m_i) = \frac{m_i}{m_i + \gamma_i}, \quad \gamma_i > 0, \tag{4}$$

with $\delta(0) = 0$, $\delta \in (0, 1)$, $\delta' > 0$, $\delta'' < 0$, and $\lim_{m_i \to \infty} \delta(m_i) = 1$. The Jeffrey-shifted posterior is $\text{Beta}(\alpha_i^{**}, \beta_i^{**})$ where:

$$\alpha_i^{**} = \alpha_i^* + \delta(m_i), \qquad \beta_i^{**} = \beta_i^*. \tag{5}$$

Player j's Jeffrey-updated perceived moments:

$$\hat{p}_i^j = \frac{\alpha_i^{**}}{\alpha_i^{**} + \beta_i^{**}}, \qquad \mu_i^j = \hat{p}_i^j u_i + (1 - \hat{p}_i^j) d_i, \tag{6}$$

$$\sigma_i^{j2} = (u_i - d_i)^2 \hat{p}_i^j (1 - \hat{p}_i^j) = (u_i - d_i)^2 \frac{(\alpha_i^* + \delta(m_i))\, \beta_i^*}{(\alpha_i^* + \delta(m_i) + \beta_i^*)^2}. \tag{7}$$

2.3 Investment, Capital Augmentation, and Preferences

Each player i divides budget $W_i > 0$ across three uses. First, marketing spend $m_i \geq 0$ is paid upfront, leaving an effective investable budget $W_i^{\text{eff}} = W_i - m_i$. Second, a fraction $(1 - \alpha_i)$ of W_i^{eff} is deployed into player i's own process. Third, the remaining fraction $\alpha_i \in [0, 1]$ is invested cross into player j's process. The budget identity is therefore:

$$W_i = m_i + (1 - \alpha_i)\, W_i^{\text{eff}} + \alpha_i\, W_i^{\text{eff}}, \tag{8}$$

with $W_i^{\text{eff}} = W_i - m_i$ substituted throughout.

Cross-investment is productive: when player j places $\alpha_j W_j^{\text{eff}}$ into player i's process, that capital is genuinely deployed and earns the same per-unit return r_i^t as player i's own capital. Player i retains a performance fee $f \in (0, 1)$ on the *gross* return earned by this incoming capital. Player i's total effective capital base — own investment plus fee-adjusted incoming capital — is therefore:

$$\tilde{A}_i = (1 - \alpha_i)\, W_i^{\text{eff}} + f\, \alpha_j\, W_j^{\text{eff}}. \tag{9}$$

The first term is player i's own deployment; the second is the fee income entitlement on the capital placed by player j.

Conversely, when player i cross-invests $\alpha_i W_i^{\text{eff}}$ into player j's process, player j retains the fee f on that capital's gross return. Player i therefore receives only the net-of-fee fraction:

$$\tilde{B}_i = \alpha_i\, W_i^{\text{eff}}\, (1 - f). \tag{10}$$

Terminal wealth is the sum of returns on each leg:

$$W_i(T) = r_i^t\,\tilde{A}_i + r_j^t\,\tilde{B}_i = r_i^t \underbrace{\big((1-\alpha_i)\,W_i^{\text{eff}} + f\,\alpha_j\,W_j^{\text{eff}}\big)}_{\tilde{A}_i} + r_j^t \underbrace{\alpha_i\,W_i^{\text{eff}}(1-f)}_{\tilde{B}_i}. \tag{11}$$

The first term is the return on player i's own process, earned on both own capital and the fee entitlement from incoming capital. The second term is the net-of-fee return on the cross-investment in player j's process.

Players maximise mean-variance utility $\mathcal{V}_i = \mathbb{E}[W_i(T)] - (\rho_i/2)\mathrm{Var}[W_i(T)]$, $\rho_i > 0$, using true moments (μ_i, σ_i^2) for their own process and Jeffrey-shifted beliefs (μ_j^i, σ_j^{i2}) for the other's process.

2.4 Utility Maximisation and Optimal Decisions

Each player i chooses marketing spend $m_i \geq 0$ and cross-investment share $\alpha_i \in [0,1]$ to maximise mean-variance utility:

$$\max_{m_i,\,\alpha_i}\ \mathcal{V}_i = \mathbb{E}[W_i(T)] - \frac{\rho_i}{2}\mathrm{Var}[W_i(T)], \tag{12}$$

where $W_i(T)$ is given by (11), using true moments (μ_i, σ_i^2) for the own process and Jeffrey-shifted beliefs (μ_j^i, σ_j^{i2}) for the other's process. The two choice variables enter differently: α_i governs the allocation of the investable budget W_i^{eff} between own and cross investment, while m_i reduces W_i^{eff} directly but raises the fee income from incoming capital by shifting player j's beliefs.

Optimal cross-investment share α_i^*. Expanding $\mathbb{E}[W_i(T)]$ and $\mathrm{Var}[W_i(T)]$ using (11) and the independence of r_i^t and r_j^t:

$$\mathbb{E}[W_i(T)] = \mu_i\,\tilde{A}_i + \mu_j^i\,\tilde{B}_i, \tag{13}$$

$$\mathrm{Var}[W_i(T)] = \sigma_i^2\,\tilde{A}_i^2 + \sigma_j^{i2}\,\tilde{B}_i^2. \tag{14}$$

Taking the first-order condition with respect to α_i and solving, the optimal cross-investment share is:

$$\alpha_i^* = \frac{1}{W_i^{\text{eff}}} \cdot \frac{\mu_j^i(1-f) - \rho_i\,\sigma_j^{i2}(1-f)^2 W_i^{\text{eff}}\,\alpha_i + \rho_i\,\sigma_i^2 f\,\alpha_j W_j^{\text{eff}} \cdot (1-\alpha_i)W_i^{\text{eff}}}{\rho_i\big(\sigma_i^2 + \sigma_j^{i2}(1-f)^2\big)}. \tag{15}$$

In the symmetric case $\sigma_i^{i2} = \sigma_j^{i2}$ this simplifies to a standard mean-variance ratio: the cross-investment share rises with the perceived Sharpe ratio of player j's process (net of fee) and falls with risk aversion ρ_i. Crucially, α_i^* depends on the Jeffrey-shifted beliefs (μ_j^i, σ_j^{i2}) and hence on m_j — player j's marketing spend — through (6)–(7).

Optimal marketing spend m_i^*. Substituting α_i^* back into $\mathcal{V}_i$ and differentiating with respect to m_i, the first-order condition for interior marketing spend equates marginal benefit to marginal cost:

$$\underbrace{f\,\alpha_j^*\,W_j^{\text{eff}}\,\mu_i\cdot\frac{\partial\hat{p}_i^j}{\partial m_i}(u_i-d_i)}_{\text{marginal fee income from belief shift}}=\underbrace{\mu_i\,(1-\alpha_i^*)+\mu_j^i\,\alpha_i^*(1-f)}_{\text{marginal cost: foregone return}}. \tag{16}$$

The left-hand side is the marginal benefit: an infinitesimal increase in m_i shifts $\hat{p}_i^j$ upward at rate $\partial\hat{p}_i^j/\partial m_i = \delta'(m_i)\,\beta_i^{**}/S^{**2}$ (from (5)), raising the perceived attractiveness of player i's process to player j, and hence increasing the incoming cross-investment that earns fee income f. The right-hand side is the marginal cost: each unit of m_i reduces W_i^{eff} by one, forgoing the return that capital would have earned in the own and cross positions.

The solution $m_i^* > 0$ is strictly interior for all finite n, because $\partial\hat{p}_i^j/\partial m_i \sim 1/n > 0$ keeps the marginal benefit positive while the marginal cost is bounded away from zero. The key feature of m_i^* for what follows is its sensitivity to the state of beliefs:

$$\frac{\partial m_i^*}{\partial\hat{p}_i^j} > 0, \tag{17}$$

since a higher perceived quality $\hat{p}_i^j$ raises the fee income on incoming capital, increasing the return to further marketing. It is precisely this responsiveness — the fact that optimal marketing spend tracks the belief state — that generates mean reversion in conditional variance, as shown in Section 3.

3 Structural GARCH and the Role of Jeffrey Conditioning

3.1 Belief Dynamics and Variance Propagation

Index periods $t = 1, 2, \ldots$ The return surprise:

$$\epsilon_i^t = r_i^t - \mu_i = \begin{cases} (1-p_i)(u_i-d_i) & \text{up move, prob. } p_i, \\ -p_i(u_i-d_i) & \text{down move, prob. } 1-p_i. \end{cases} \tag{18}$$

Derivation of the posterior mean update. At the start of period t, player j's Jeffrey-shifted posterior on p_i is $\text{Beta}(\alpha_i^{**,t-1}, \beta_i^{**,t-1})$ as given by (5), with total pseudo-count $S_t = \alpha_i^{**,t-1} + \beta_i^{**,t-1}$ and posterior mean $\hat{p}_i^{j,t-1} = \alpha_i^{**,t-1}/S_t$. After observing period

t's return, the posterior updates by conjugacy: an up-move increments α_i^{**} by one, a down-move increments β_i^{**} by one, and S_t increases to S_t+1 in either case. The resulting shift in posterior mean is:

$$\begin{aligned}
\text{up move:} \quad & \frac{\alpha_i^{**,t-1}+1}{S_t+1} - \frac{\alpha_i^{**,t-1}}{S_t} = \frac{\beta_i^{**,t-1}}{S_t(S_t+1)}, \\
\text{down move:} \quad & \frac{\alpha_i^{**,t-1}}{S_t+1} - \frac{\alpha_i^{**,t-1}}{S_t} = \frac{-\alpha_i^{**,t-1}}{S_t(S_t+1)}.
\end{aligned}$$

Combining both cases via indicator functions:

$$\Delta \hat{p}_i^{j,t} = \frac{\mathbf{1}_{\text{up}}\, \beta_i^{**,t-1} - \mathbf{1}_{\text{down}}\, \alpha_i^{**,t-1}}{S_t(S_t+1)}, \tag{19}$$

which follows directly from (3) and (5).

Approximation in terms of the return surprise. To connect (19) to the return surprise ϵ_i^t, note that $\mathbf{1}_{\text{up}} = \epsilon_i^t/(u_i-d_i)(1-p_i)$ and $\mathbf{1}_{\text{down}} = -\epsilon_i^t/(u_i-d_i)p_i$ in expectation. Weighting the numerator of (19) by these expectation factors and collecting terms yields:

$$\Delta \hat{p}_i^{j,t} \approx \frac{\bar{C}_t\, \epsilon_i^t}{(u_i-d_i)\, S_t(S_t+1)}, \tag{20}$$

where

$$\bar{C}_t = \frac{p_i\, \beta_i^{**,t-1}}{1-p_i} + \frac{(1-p_i)\, \alpha_i^{**,t-1}}{p_i} \tag{21}$$

is a positive weighting coefficient that captures the asymmetric sensitivity of the posterior mean to up- versus down-moves, governed by the current Jeffrey-shifted parameters $(\alpha_i^{**,t-1}, \beta_i^{**,t-1})$. Since $\mathbb{E}[\epsilon_i^t \mid \mathcal{F}_t] = 0$, the approximation (20) is exact in expectation.

The belief update propagates into realised portfolio variance $\mathcal{R}_i = \mathrm{Var}[W_i(T)]$ through four channels: the drift channel, the volatility channel, the marketing adjustment channel, and the capital augmentation channel. Let Λ_t denote the total sensitivity of realised variance to the current belief level. Taking the second-order expansion of $\mathbb{E}[\mathcal{R}_{i,t+1} \mid \mathcal{F}_t]$ and using $\mathbb{E}[\epsilon_i^t \mid \mathcal{F}_t] = 0$ (so the linear term vanishes), one obtains:

Theorem 1 (Structural GARCH(1,1))**.** *Under Jeffrey conditioning with optimal marketing, the conditional variance satisfies:*

$$\mathbb{E}[\mathcal{R}_{i,t+1} \mid \mathcal{F}_t] = \underbrace{\omega_i^t}_{\textit{baseline}} + \underbrace{\alpha_i^{\text{GARCH}}\, (\epsilon_i^t)^2}_{\textit{ARCH}} + \underbrace{\beta_i^{\text{GARCH}}\, \mathcal{R}_{i,t}}_{\textit{persistence}}, \tag{22}$$

with structural coefficients derived from model primitives:

$$\omega_i^t = \frac{\kappa_t}{S_t}\bar{\mathcal{R}}_i + \underline{\sigma}_i^2, \tag{23}$$

$$\alpha_i^{\mathrm{GARCH}} = \frac{\bar{C}_t^2}{2S_t^2(S_t+1)^2}\frac{\partial \Lambda_t}{\partial \hat{p}_t}, \tag{24}$$

$$\beta_i^{\mathrm{GARCH}} = 1 - \frac{\partial m_i^{*,t}}{\partial \hat{p}_i^{j,t}} \cdot \frac{\partial \hat{p}_i^{j,t}}{\partial \mathcal{R}_{i,t}}, \tag{25}$$

where $\underline{\sigma}_i^2 = p_i(1-p_i)(u_i-d_i)^2$ is the irreducible process variance floor, κ_t is the strength of the marketing mean-reversion adjustment, and Λ_t is derived explicitly in Rana (2025).

A GJR-GARCH extension adds an endogenous leverage term:

$$\mathbb{E}[\mathcal{R}_{i,t+1} \mid \mathcal{F}_t] = \omega_i^t + \alpha_i^{\mathrm{GARCH}}(\epsilon_i^t)^2 + \gamma_i^{\mathrm{GARCH}}(\epsilon_i^t)^2 \mathbf{1}_{\epsilon_i^t<0} + \beta_i^{\mathrm{GARCH}}\mathcal{R}_{i,t}, \tag{26}$$

where the sign of $\gamma_i^{\mathrm{GARCH}}$ depends on $\hat{p}_t \gtrless 1/2$, making the leverage effect endogenous and regime-dependent rather than parametrically imposed.

3.2 The Central Result: Jeffrey Conditioning Restores Stationarity

Proposition 1 (Jeffrey vs. Bayesian)**.** *(i)* ***Under Jeffrey conditioning with optimal marketing:*** *The persistence coefficient satisfies $\beta_i^{\mathrm{GARCH}} < 1$, so the GARCH process (22) is covariance-stationary whenever $\alpha_i^{\mathrm{GARCH}} + \beta_i^{\mathrm{GARCH}} < 1$. The mean-reversion force is:*

$$1 - \beta_i^{\mathrm{GARCH}} = \frac{\partial m_i^{*,t}}{\partial \hat{p}_i^{j,t}} \cdot \frac{\partial \hat{p}_i^{j,t}}{\partial \mathcal{R}_{i,t}} > 0. \tag{27}$$

(ii) ***Under pure Bayesian updating ($m_i \equiv 0$):*** *The marketing adjustment channel vanishes entirely. Therefore:*

$$\beta_i^{\mathrm{GARCH}} = 1, \tag{28}$$

and the conditional variance follows an IGARCH process — a unit-root process with no covariance stationarity.

Proof. Part (i): Under Jeffrey conditioning, player i has a strictly interior optimal marketing spend $m_i^* > 0$ for all finite n. The derivative $\partial m_i^* / \partial \hat{p}_i^j > 0$ because higher perceived quality raises the marginal value of attracting cross-investment (higher fee income). The derivative $\partial \hat{p}_i^j / \partial \mathcal{R}_i \neq 0$ because higher realised variance reduces the attractiveness of cross-investment, reducing α_j^* and hence the fee income that makes marketing profitable. The product is strictly positive, giving $1 - \beta_i^{\text{GARCH}} > 0$.

Part (ii): Under pure Bayesian updating, soft evidence has no mechanism to shift beliefs — only hard signals with well-defined likelihoods can update the posterior. Marketing therefore cannot influence beliefs and $m_i \equiv 0$ is the only feasible choice. With $m_i^{*,t} \equiv 0$ for all t, the partial derivative $\partial m_i^{*,t} / \partial \hat{p}_i^{j,t} = 0$ identically, and equation (25) reduces immediately to $\beta_i^{\text{GARCH}} = 1$. □

Remark 1. *The result is structural, not parametric. Stationarity in* (22) *does not depend on the values of* ρ_i*,* f*,* u_i*,* d_i*, or the prior parameters. It depends only on whether the belief-updating rule permits soft evidence — and hence optimal marketing — to exist as an endogenous strategic variable. Jeffrey conditioning permits it; Bayesian updating does not.*

3.3 Properties of the Structural Coefficients

Each coefficient has a precise economic source and a clean comparative static in n:

ARCH coefficient $\alpha_i^{\text{GARCH}} \sim 1/n^2$*:* Longer history anchors beliefs, making individual surprises less capable of propagating into future variance. As $n \to \infty$, the ARCH effect vanishes.

Persistence $\beta_i^{\text{GARCH}} \to 1$ *as* $n \to \infty$*:* Marketing spend decays at rate $1/\sqrt{n}$ as accumulated history weakens the effectiveness of the lie (see Proposition 2 below). The marketing adjustment accordingly weakens, and β_i^{GARCH} approaches unity — though the stationarity condition is preserved for all finite n since α_i^{GARCH} decays faster ($1/n^2$) than $1 - \beta_i^{\text{GARCH}}$ ($1/\sqrt{n}$).

GJR asymmetry $\gamma_i^{\text{GARCH}} \sim 1/n^2$*:* The endogenous leverage effect is strongest in short relationships and disappears as history accumulates.

Baseline $\omega_i^t \geq \underline{\sigma}_i^2 > 0$*:* Bounded below by the irreducible process variance for all n, preventing variance from collapsing to zero even as all dynamic effects vanish.

Corollary 1 (Stationarity Condition)**.** *The process* (22) *is covariance-stationary if and only if:*

$$\frac{\bar{C}_t^2}{2S_t^2(S_t+1)^2} \frac{\partial \Lambda_t}{\partial \hat{p}_t} < \frac{\partial m_i^{*,t}}{\partial \hat{p}_i^{j,t}} \cdot \frac{\partial \hat{p}_i^{j,t}}{\partial \mathcal{R}_{i,t}}, \tag{29}$$

i.e., the ARCH surprise effect is dominated by the marketing mean-reversion force. This holds whenever fee income $f \cdot \mu_i \cdot W_j$ *is sufficiently large relative to belief sensitivity to surprises.*

Table 1 summarises the regime dependence.

Table 1: GARCH coefficient behaviour across regimes. All parameters derived from model primitives.

Coefficient	Jeffrey, finite n	Bayesian ($m_i \equiv 0$)	Jeffrey, $n \to \infty$
ω_i^t	$> \underline{\sigma}_i^2$, time-varying	$= \underline{\sigma}_i^2$, fixed	$\to \underline{\sigma}_i^2$
$\alpha_i^{\mathrm{GARCH}}$	> 0, decays $\sim 1/n^2$	> 0 (same)	$\to 0$
β_i^{GARCH}	< 1 **(stationary)**	$= 1$ **(IGARCH)**	$\to 1$
$\gamma_i^{\mathrm{GARCH}}$	endogenous sign	endogenous sign	$\to 0$
Stationarity	**Yes**	**No**	Marginal

3.4 The Lie Decays Asymptotically

Proposition 2 (Asymptotic Dominance of Truth)**.** *As $n \to \infty$:*

$$m_i^* \to 0, \qquad \hat{p}_i^j \to p_i, \qquad \sigma_i^{j2} \to \sigma_i^2. \tag{30}$$

Optimal marketing spend decays at rate $1/\sqrt{n}$. Perceived beliefs and perceived volatility converge to their true values at rate $1/n$.

Proof. The marginal belief shift satisfies $\Pi = \partial \hat{p}_i^j / \partial m_i \sim \beta_i^{**}/S^2 \sim 1/n$. Since the marginal benefit of marketing (fee income shifted by belief manipulation) contains Π as a factor while marginal cost (foregone own-process return) is bounded away from zero, the optimal m_i^* decays at rate $1/\sqrt{n}$ and reaches zero at finite threshold n^*. Thereafter $\delta(m_i^*) \to 0$ and $\hat{p}_i^j \to \alpha_i^*/S \to p_i$ by the law of large numbers. $\square$

Remark 2. *This result holds regardless of the marketing budget or the parameters of the model. Truth — encoded in the public history of returns — inevitably dominates strategic distortion. The lie is a short-run phenomenon: most effective when relationships are young and history is short, progressively weaker as the relationship matures, and ultimately irrelevant in the long run.*

4 Discussion

Connection to RBE literature. Kurz (1994) and Kurz and Motolese (2001) derive GARCH from the aggregate distribution of heterogeneous beliefs in competitive general equilibrium. Our mechanism is bilateral and strategic rather than distributional and competitive. Agents in RBE ignore their effect on prices; our players explicitly account for how marketing shifts the other's beliefs and hence allocation. The soft evidence channel through which marketing operates is absent from RBE, which relies on rational but diverse models of a complex economy.

Connection to Kyle (1985). Kyle (1985) shows that informed trading produces a martingale price with constant volatility, converging to full information revelation by the terminal date. Our model also achieves asymptotic informational efficiency (Proposition 2), but through an opposite mechanism: Kyle's insider conceals information and profits from price impact; our player embellishes beliefs and profits from fee income. Neither generates stationary GARCH — Kyle produces constant volatility, Bayesian updating produces IGARCH. Jeffrey conditioning is the ingredient that restores stationarity in our setting.

Connection to Bayesian persuasion. Kamenica and Gentzkow (2011) study a sender who designs signals to persuade a Bayesian receiver. Three distinctions apply here: belief updating is via Jeffrey rather than Bayes; communication is costly and funded from the investment budget (making the interior optimum for m_i^* depend on economic primitives rather than arising from signal design); and both players simultaneously act as senders and receivers within a Stackelberg game rather than a one-sided persuasion game.

5 Conclusion

We have shown that the choice between Bayesian and Jeffrey updating has structural consequences for volatility dynamics in a strategic investment model. Under Bayesian updating, the conditional portfolio variance follows IGARCH with unit-root persistence. Under Jeffrey conditioning, optimal marketing spend introduces a mean-reversion force that converts IGARCH into stationary GARCH(1,1). All GARCH parameters are derived as explicit functions of model primitives. The GJR leverage asymmetry is endogenous and regime-dependent. Truth dominates the lie asymptotically: optimal marketing decays at rate $1/\sqrt{n}$ and beliefs converge to truth at rate $1/n$ as the shared public history accumulates.

The result identifies Jeffrey conditioning — and the strategic marketing it enables — as the structural source of mean reversion in conditional variance. This provides a microfoundation for empirically observed GARCH stationarity that does not require exogenous shocks to volatility or ad hoc persistence assumptions.

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